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2. When multiplying or dividing fractions, there is no requirement for a common denominator.
3. Let's look at an example. What is two thirds plus three sevenths?
4. We will rewrite both fractions to have twenty-one in the denominator.
5. For the first fraction, multiply the top and the bottom by seven. We haven't finished this step, which explains the space after the plus sign.
6. For the second fraction, multiply the top and the bottom by three.
7. Simplifying each numerator and denominator separately, we have fourteen over twenty-one plus nine over twenty-one.
8. Now that we have the same denominator, we only add numerators. Since fourteen plus nine is twenty-three, our single fraction is twenty-three over twenty-one. We prefer improper fractions over mixed fractions.

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Example. $\frac{1}{10} + \frac{1}{6}$.

1. Let's look at another example. What is one tenth plus one sixth?
2. We could rewrite both fractions to have sixty in the denominator, but ten and six both have a common factor of two, so will rewrite both fractions to have thirty in the denominator.
3. For the first fraction, multiply the top and the bottom by three.
4. For the second fraction, multiply the top and the bottom by five.
5. Simplifying each numerator and denominator separately, we have three over thirty plus five over thirty.
6. Now that we have the same denominator, we only add numerators. Since three plus five is eight, our single fraction is eight over thirty.
7. We can simplify this fraction by cancelling a factor of two on top and bottom, leaving us with four over fifteen.

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1. Let's look at another example. What is three fourteenths plus one seventh?
2. Since fourteen times seven is ninety-eight, we could rewrite both fractions to have ninety-eight in the denominator, but we can actually just make both fractions have fourteen in the denominator.
3. Copy the first fraction as is, with no changes.
4. For the second fraction, multiply the top and the bottom by two.
5. Unlike in previous examples, we didn't even change how both fractions are written. We only rewrote the second fraction.
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